INCH-POUND

MIL-I-24768/11
25 February 1992

SUPERSEDING
MIL-P-79C
15 June 1961

MILITARY SPECIFICATION SHEET

INSULATION, PLASTIC, LAMINATED, THERMOSETTING, PAPER-BASE, PHENOLIC-RESIN (PBG)

This specification is approved for use by all departments and agencies of the Department of Defense.

The requirements for acquiring the insulation material described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-I-24768.

REQUIREMENTS:

Construction: The material shall consist of a cellulosic paper substrate properly impregnated and bonded with a phenolic-resin matrix or binder and processed to conform to this specification.

Intended use: The laminated material covered by this specification is a moisture and humidity resistant, plastic laminate for usual electrical applications.

Forms: Sheets, rolled tubes, molded tubes, and rods are recommended forms of laminated material covered by this specification.

Temperature indices: The laminated material covered by this specification has a mechanical strength test index of 130 and an electrical strength test index of 130 for thickness up to 0.062 inch and a mechanical strength test index of 140 and an electrical strength test index of 140 for thicknesses greater than 0.062 inch.

Dimensions and part numbers:

Thickness of sheets: The thickness of laminated sheets, permissible variations, and the applicable part number shall be as specified in table I.

AMSC N/A FSC 5970

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TABLE I. *Thickness of laminated sheets.*

Part number	Nominal thickness (inches)	Permissible variations (± inch)
M24768/11-S-1	0.010	0.002
M24768/11-S-2	0.015	0.025
M24768/11-S-3	0.020	0.003
M24768/11-S-4	0.025	0.0035
M24768/11-S-5	0.031	0.0035
M24768/11-S-6	0.047	0.0045
M24768/11-S-7	0.062	0.005
M24768/11-S-8	0.094	0.007
M24768/11-S-9	0.125	0.008
M24768/11-S-10	0.156	0.009
M24768/11-S-11	0.188	0.010
M24768/11-S-12	0.219	0.011
M24768/11-S-13	0.250	0.012
M24768/11-S-14	0.312	0.014
M24768/11-S-15	0.375	0.017
M24768/11-S-16	0.438	0.019
M24768/11-S-17	0.500	0.021
M24768/11-S-18	0.625	0.024
M24768/11-S-19	0.750	0.027
M24768/11-S-20	0.875	0.030
M24768/11-S-21	1.000	0.033
M24768/11-S-22	1.125	0.035
M24768/11-S-23	1.250	0.037
M24768/11-S-24	1.375	0.039
M24768/11-S-25	1.500	0.041
M24768/11-S-26	1.625	0.043
M24768/11-S-27	1.750	0.045
M24768/11-S-28	1.875	0.047
M24768/11-S-29	2.000	0.049
M24768/11-S-30	2.125	0.052
M24768/11-S-31	2.500	0.056
M24768/11-S-32	2.750	0.061
M24768/11-S-33	3.000	0.065
M24768/11-S-34	3.500	0.073
M24768/11-S-35	4.000	0.082
M24768/11-S-36	4.500	0.090
M24768/11-S-37	5.000	0.095
M24768/11-S-38	5.500	0.105
M24768/11-S-39	6.000	0.115
M24768/11-S-40	6.500	0.120
M24768/11-S-41	7.000	0.130
M24768/11-S-42	7.500	0.140
M24768/11-S-43	8.000	0.145
M24768/11-S-44	8.500	0.155
M24768/11-S-45	9.000	0.160
M24768/11-S-46	9.500	0.170
M24768/11-S-47	10.000	0.180
M24768/11-S-48	_[1]	_[2]

See footnotes on next page.

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[1]For part numbers of sheets not listed in this table, the thickness shall be included after the part number. Example of this for a 0.8125 inch thick sheet shall be: M24768/11-S-48-0.8125.
[2]On sheets of nominal thickness not listed in this table, the permissible variations shall be the same as for the next greater thickness.

Diameter of rolled round tubes: The range of sizes for rolled round tubes shall be as specified in table II. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a rolled round tube with an inside diameter of 0.125 inch and an outside diameter of 0.156 inch is: M24768/11-TRR-0.125/0.156. The wall thickness tolerances for rolled tubes with an inside diameter up to 4 inches shall be as specified in table III.

TABLE II. *Range of sizes for rolled round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	48.000	0.145	50.000	0.010	1.000	1/2

TABLE III. *Rolled round tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)
0.010 to 0.015	0.003
0.016 to 0.030	0.005
0.31 to 0.061	0.006
0.062 to 0.124	0.007
0.125 to 0.249	0.009
0.250 to 0.500	0.011

Diameter of molded round tubes: The range of sizes of molded tubes shall be as specified in table IV. The inside diameter and outside diameter shall be included in the part number. An example of a part number for a molded round tube with an inside diameter of 0.125 inch and an outside diameter of 0.250 inch is: M24768/11- TMR-0.125/0.250. The wall thickness tolerances for molded round tubes with an inside diameter up to 3.875 inches shall be as specified in table V.

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TABLE IV. *Range of sizes for molded round tubes.*

Inside diameter (inches)		Outside diameter (inches)		Wall thickness (inch)		Maximum ratio of wall thickness to inside diameter
Minimum	Maximum	Minimum	Maximum	Minimum	Maximum	
0.125	3.875	0.250	4.000	0.062	1.000	1/2

TABLE V. *Molded round tube wall thickness tolerances.*

Wall thickness (inch)	Tolerances (± inch)		
	Inside diameter (inches)		
	0.125–0.250	0.251–0.500	0.501–Max.
0.031 to 0.061	0.008	0.008	0.008
0.062 to 0.125	0.011	0.011	0.011
0.126 to 0.250		0.015	0.011
0.251 to 0.500			0.013

Diameter of rods: The range of sizes and tolerances for rods shall be as specified in table VI. The outside diameter of the rod shall be included in the part number. An example of a part number for a rod with an outside diameter of 0.156 inch is: M24768/11-R-0.156.

TABLE VI. *Size and tolerances for rods.*

Diameter (inches)	Tolerances (± inch)
0.125 to 1.999	0.005
2.000	0.008

Performance requirements: Performance requirements for laminated materials shall be as specified in tables VII, VIII, IX, X, and XI.

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The following performance requirements do not apply to this plastic laminate:

a. Surface resistance
b. Volume resistivity
c. Arc resistance
d. Flammability
e. Thermal endurance
f. Tracking resistance
g. Silicone content.

Testing: The laminated material covered by this specification shall be tested under first article inspection requirements.

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TABLE VII. *Performance requirements for sheets.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Dielectric breakdown voltage[2]	A D-48/50	4.6.4	Min. kV	40.0 5.0	40.0 5.0	40.0 5.0	40.0 5.0	40.0 5.0	40.0 5.0	40.0 5.0	40.0 5.0	40.0 5.0	25.0 3.0
Dielectric constant	A D-24/23	4.6.6	Max.	5.50 6.30	5.50 6.20	5.50 6.10	5.50 6.00	5.50 6.00	5.50 6.00	5.50 6.00	5.50 6.00	5.50 6.00	5.50 6.00
Dissipation factor	A D-24/23	4.6.6	Max.	0.045 0.053	0.045 0.052	0.045 0.051	0.045 0.050	0.045 0.050	0.045 0.050	0.045 0.050	0.045 0.050	0.045 0.050	0.045 0.050
Volume resistivity	C-96/35/90	4.6.7	Min. megohm cm										
Surface resistance	C-96/35/90	4.6.8	Min. megohm										
Arc resistance	D-48/50	4.6.9	Min. sec.										
Tracking resistance	A	4.6.10	Min. minutes										
Impact strength[2] Lengthwise: Crosswise:	E-48/50	4.6.11	Min. ft-lbs. per inch				0.40 0.35	0.40 0.35	0.40 0.35	0.40 0.35	0.40 0.35	0.40 0.35	0.40 0.35
Flexural strength Lengthwise: Crosswise:	A	4.6.12	Min. psi	15,000 13,000	15,000 14,000	15,000 14,000	15,000 14,000	15,000 14,000	15,000 14,000	15,000 14,000	15,000 14,000	13,500 12,500	13,500 12,500

See footnotes at end of table.

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TABLE VII. *Performance requirements for sheets – Continued.*

Requirement	Cond.	Test para.	Unit	Thickness (inches)[1]									
				0.031	0.062	0.094	0.125	0.188	0.250	0.500	0.750	1.000	1.001 – Max.
Thermal endurance Flexural strength:		4.6.13	Min. psi										
Bonding strength[2]	A		Min. pounds							800	800	800	800
	D-48/50	4.6.15								600	600	600	600
Water absorption	D_1-24/23	4.6.17	Max. percent	3.10	2.00	1.60	1.30	1.00	0.85	0.55	0.50	0.45	0.45
Silicone content	E-168/185	4.6.18	Max. ppm										
Flame resistance													
Ignition time:	A	4.6.19	Min. sec.										
Burning time:			Max. sec.										

[1]For intermediate thicknesses, the value for the next smaller thickness shall apply.
[2]Maximum thickness tested shall be 2.000 inches.

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TABLE VIII. *Performance requirements for rolled tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)				
					0.031–0.047	0.048–0.062	0.063–0.094	0.095–0.125	0.126–0.188
Dielectric strength	A	4.6.5	Min. volts per mil	0.125–0.499	310	310	290	290	200
				0.500–8.000	400	400	290	290	200
	D-24/23			0.125–0.499	170	170	190	190	
				0.500–8.000	250	250	250	250	
Compressive strength[1]	A	4.6.14	Min. psi	0.125–0.499	10,000	10,000	10,000	10,000	10,000
				0.500–8.000			13,000	13,000	13,000
Specific gravity	A	4.6.16	Min.	0.125–0.499	1.10	1.10	1.10	1.10	1.10
				0.500–8.000	1.12	1.12	1.12	1.12	1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125–0.499	6.0	6.0	3.6	2.5	2.0
				0.500–8.000	6.0	6.0	3.0	2.5	2.0

TABLE VIII. *Performance requirements for rolled tubes – Continued.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)				
					0.189–0.250	0.251–0.375	0.376–0.500	0.501–0.750	0.751–1.000
Dielectric strength	A	4.6.5	Min. volts per mil	0.125–0.499	200				
				0.500–8.000	200	145	145	120	105
	D-24/23			0.125–0.499					
				0.500–8.000					
Compressive strength[1]	A	4.6.14	Min. psi	0.125–0.499	10,000	10,000	10,000	10,000	10,000
				0.500–8.000	13,000	13,000	13,000	13,000	13,000
Specific gravity	A	4.6.16	Min.	0.125–0.499	1.10	1.10	1.10	1.10	1.10
				0.500–8.000	1.12	1.12	1.12	1.12	1.12
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125–0.499	1.4	1.2			
				0.500–8.000	1.4	1.2	1.0	0.9	0.9

[1]Values apply to inside diameters of 0.250 inch and over and to outside diameters of 2.000 inches and less.

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TABLE IX. *Performance requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)						
					0.062–0.093	0.094–0.124	0.125–0.187	0.188–0.249	0.250–0.499	0.500–0.999	1.000
Compressive strength	A	4.6.14	Min. psi	0.250 to 3.875	18,000	18,000	18,000	18,000	18,000	18,000	18,000
Specific gravity	A	4.6.16	Min.	0.125 to 3.875	1.25	1.25	1.25	1.25	1.25	1.25	1.25
Water absorption	D_1-24/23	4.6.17	Max. percent	0.125 to 3.875	2.0	1.8	1.6	1.3	1.2	1.0	0.9

TABLE X. *Dielectric strength requirements for molded tubes.*

Requirement	Cond.	Test para.	Unit	Inside diameter (inches)	Wall thickness (inch)			
					0.062	0.063–0.125	0.126–0.250	0.251–0.500
Dielectric strength	A	4.6.5	Min. volts per mil	0.125–3.875	300	220	150	110

TABLE XI. *Performance requirements for rods.*

Requirement	Cond.	Test para.	Unit	Diameter (inches)				
				0.125–0.249	0.250–0.499	0.500–0.999	1.000	1.001–2.000
Flextural strength	A	4.6.12	Min. psi	15,000	15,000	15,000	15,000	
Compressive strength	A	4.6.14	Min. psi	20,000	20,000	20,000	20,000	20,000
Specific gravity	A	4.6.16	Min.	1.30	1.30	1.30	1.30	1.30
Water absorption	D_1-24/23	4.6.17	Max. percent	2.5	1.5	1.0	1.0	1.3

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Custodians:
Army – ER
Navy – SH
Air Force – 20

Review Activities:
Army – MI, AR, EA
Navy – EC, OS
Air Force – 85

User Activities:
Army – ME
Navy – MC, AS

Preparing Activity
Navy – SH
(Project 5970-1015-11)

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